

Chapter 8

Basic Quantities

a) $K=8$, $H=128$, $L=80$ + 50, per token
KV cache size = $2 \cdot (8 \cdot 128 \cdot 80) \cdot (1) \text{ for int8}$
 $\approx 164 \text{ KB}$

a) $B=32$, $T=8192$, int8 quantization.

Memory_{KV} = $32 \cdot 8192 \cdot 164 \approx 43 \text{ GB}$

Params Memory = $70 \text{ GB} = 70 \text{ GB}$,

total = 113 GB .

Assuming TPU5e chips, we need at least $\frac{113}{16} = 7.06 \approx 8 \text{ chips}$

so 4×2 slice

a3) Latency is given by = $\frac{(B \times KV) + \text{Param}}{\text{HBM/s}}$

(batch is small, so we assume HBM/s)

that we are
memory bound
(MUB)

$$= \frac{43^{GB} + 70GB}{8.2 \text{ TB/s} \times 8}$$

$$\approx 17.2 \text{ ms}$$

$$\text{Throughput} = (\text{latency})^{-1} \cdot B = \frac{1}{17.2} \cdot 32 = 1.86 \text{ tok / ms}$$

$$\text{Per chip Throughput} = \frac{1.86}{8} = 0.231 \text{ tokens/ms/chip}$$

$$\boxed{231 \text{ tokens/sec/chip}}$$

Since we are decoding, we are in generation, so our scaling strategy is TP. So, we are complete here if

$$V > M_V \cdot \frac{F}{2550} \quad \text{ie } V > 2 \cdot \frac{28672}{2550}$$

$$\approx 223$$

For $4 \times 2 = 8$ and $4 \times 4 = 16$, these are not relevant so we know we are complete here in TP. (more specifically, we know that $\# \text{ of } \text{TP} > 223$, so it follows that $\text{time} > 76\%$ so we have small B)

For 4×4 , we have latency = $\frac{113 \text{ GB}}{8020 \text{ B/s}} = 17.2 \text{ ms} = \boxed{8.6 \text{ ms}}$

total
and throughput go from $\boxed{1860 \text{ tokens/second to } 3720 \text{ tokens/second.}}$

Thinking About Throughput

- (Q1) For TPVSe with bf16 all, we have 240 (from chapter 2).
 if we store 2 int8, we have $240 = 120$ because
 HBM is twice as fast. if we store and compute int8, then
 both HLOs and HBM's reduce by 2, so no change. is 240.

(Q2) ~~If we say the size is not fixed, then we say a fixed size~~

- 700 of model maxims. So, for bf16, we have
 $bf16 \ 140GB / 16 = 8.75$, so we'd need a slice of 4×4
 $int8 \ 700GB / 16 = 4.4$, so we'd need a slice of 4×2
 $int4 \ 350GB / 16 = 2.18$ so a ~~4x2~~ slice would suffice

Again

2x2

~~So the HLO is not fixed, we can have a fixed size, so that is 16 HLO, 8192~~

- For ~~bf16~~ bf16, $KV = 16 \times 2 \cdot 8192 = 2.7GB$ per sequence
 For int8, $KV = 2.7GB / 2 = 1.35GB$
 For int4, $KV = 0.672GB$.

- (Q3) We first compute the largest data size that fits on exclusive. We have
 for bf16, $16 \times 16 = 256GB$ in HBM, with $256 - 140 = 116GB$ for KV.
 Each seq needs ~~2.7GB~~ 2.7GB mem, so we have $\max P = \frac{116}{2.7} = 42.9$
 $= 42$

For int8, $8 \times 16 = 128$, with $128 - 70 = 58$ for KV,

and $\frac{58}{1.35} = 42$,

for int4, also 42 (everything divided by 2, so it doesn't change).

$B=42$. This is less than ~~240, 120, 60~~ (assuming bflb compute) we are mem bound

$$\begin{aligned} \text{Latency} &= \text{Params} + (42 \cdot 2.76B) \quad \text{bflb} \\ &= \text{Params} + (42 \cdot 1.356B) \quad \text{Int 8} \quad (\text{all divided by } 2) \\ &= \text{Params} + 42(0.676B) \quad \text{Int 4} \quad (\text{all divided by } 2) \end{aligned}$$

which gives $\approx 19ms$ bflb
 $\approx 19ms$ Int 8
 $\approx 19ms$ Int 4

again, if every number is divided by 2, so the time

Q4) Let $\text{Thr} = \text{Throughput}$.

$$\text{Thr}_{\text{tokens}} = (\text{latency}) \cdot B = \frac{1}{0.014s} \cdot 42 = 2210 \text{ tokens/sec}$$

$$= 138 \text{ tokens/sec/chip} \quad \text{queries/chip/sec}$$

$$\text{Thr}_{\text{queries}} = \frac{\text{Thr}_{\text{tokens}}}{\text{median doc length}} = \frac{138}{512} \approx 0.27 \text{ at bflb.}$$

$$\text{For Int 8, This is } 2210 / (8 \cdot 512) = 0.2712 =$$

$$0.54 \text{ queries/chip/sec at Int 8,}$$

Ans 4 is 1.09 queries/chips/second.

Q5) Memory for KV ~~is HBM - Param MEM~~

Q5) The available memory for KV is $HBM - Param_{MEM}$.
For all quantizations, $B = \frac{(HBM - Param_{MEM})}{KV_{size}}$. This

ratio is constant ~~bc~~

bc going from b1b

to 2t8 means dividing each value, this
equation by 2.

The question asks "what if we doubled HBM". Since this
equally influences the ratio of lecture memory for each
quantization, ~~we~~ the same property of KV is ~~the same~~
new B for each of them will ~~remain~~ be the same across quantizations.
ie B for b1b, 2t8, 2t4 remain the same, so
we have $\frac{512 - 140}{2 \cdot 8} = 37.25 \text{ GB for KV} = 137 \text{ B}$, so

$$\text{New throughput} = \frac{137}{(0.019) \cdot 32 \cdot 512} = 0.44 \text{ queries/sec/chip for b1b}$$

so we have

0.88 q/sec/chip for 2t8 and
1.76 q/sec/chip for 2t4

(b) Since we're in generation, we can only do model parallelism. So, we can have $4 \times 3 = 12$ way model parallelism.

We do have that $32 > 2B = 2 \cdot \frac{28672}{2500} = M_g \cdot \frac{F}{2500}$

We are not compute bound. But however, when we can increase γ ~~to 64~~ ^{this} ~~way~~ be we don't incur a penalty for not being compute bound because we are comm bound anyway (from chapter 7). We have

$$\gamma > \frac{F}{C \cdot (W_{HBM}/W_{MEM})} = \frac{28672}{137 \cdot (8.2e11/9.0e9)} = 23$$

So unfortunately, this is still about 23, so we ~~can't do~~ will have to eat the inefficiency.

We could reduce B to 24. Then we have $\gamma > 131$, so this can't work. We could even do $B = 18$ to get $\gamma > 65$, which means we can in fact be HBM bound and avoid FC-boundness.

If we go to an even higher size ($4 \times 4 \times 8$), then we could do ~~the~~ 2P weight stationary sharding and potentially see a better utilization of hardware.

Prefill

Q1) Prefill length = 8192 tokens; 16 v5F chips, so per chip
 $B = 512 \times 240$, so we are compute bound. So,
we have about $2^{14} \cdot 8192 \cdot 70\%$ total FLOPs for a
forward pass, so we have $\frac{1.14e15}{1.97e14 \cdot 16} = 0.36$ sec at
100% util, so

$$\text{we have } 0.36 \cdot \frac{10}{4} = \boxed{0.904 \text{ sec}}$$

Q2) Median Prefill Length = 8192

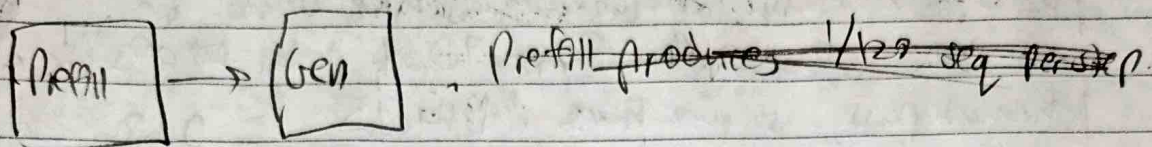
Median Decode Length = 4096

$B = 32$. One step producing 32 tokens, so about
 $\frac{32}{4096} = \frac{1}{128}$ ~~seqs~~ sequences finish every step.

Each seq uses $8192 + 4096 = 12288$. Since $\frac{1}{128}$ seqs
finish each step, we have $\frac{12288}{128} = \boxed{96 \text{ tokens}}$

enter from KV cache per step

(Q3) We are doing disaggregated serving, i.e. prefill server + generation server.



Throughput = $B \cdot \text{latency}^{-1}$. If prefill and gen are to be equally shared, then prefill needs to produce enough tokens for generation to consume. Generation consumes at $B = 32$, 32 tokens per step, if token decode = 512, then it takes

~~it takes~~ ~~about~~ ~~512~~ tokens to process a seq, so

it takes about 16 steps to process a sequence. For 16 steps, 16ms, so we have 32ms/seq. For prefill prefill produces

~~about~~ 1 seq in ~~904ms~~ 904ms, so we have

$\frac{904 \text{ ms/seq}}{32 \text{ tokens/seq}} \approx 28$, i.e. prefill time is as long to produce

a seq as generating, we need roughly $3 \times$ the prefill server as generation

Warm Problems

- Q1) A forward pass uses about 2. param tokens, so ~~for~~ ^{use}
 $= 810e9$ flops per token. With N chips, we have

$$\frac{810e9}{N \cdot 1.07e14} = \frac{0.988}{N} \text{ sec} \quad (\text{assuming 64B flops})$$

$$\boxed{\frac{4}{N} \text{ ms}}$$

If we are commo band, then (assuming 8x8 storage) we have

$$\frac{405e9}{N \cdot 8 \cdot 2e11} = \frac{494 \text{ ms}}{N} \quad (\text{if token batch size} = 1)$$

- Q2) Model params use 809 bytes = $\boxed{86B}$. $(T = \text{context})$

b) KV cache uses $240 \cdot (8 \cdot 128 \cdot 8) = \boxed{7.866B}$

c) Max ~~sequence~~ internal ~~sequence~~ size = $240 \cdot F = 240 \cdot 14336 = \boxed{3.446B}$

For sequence, we'll assume 8192 ~~context~~ ~~tokens~~ median prefll.

- Q3) 405e9 with 8x8 storage = $\frac{405GB}{16} \approx 25$, so at least 9
 $\approx 4 \times 8$ slice.

This leaves 107GB for KV, so $2 \times 126 \times 128 \times 8 \times 8192$
 $= 2.116B$ per sequence.

With 107 sparc, $B \approx 48$. Our latency is $\frac{405 + (48)2.011}{32 \cdot 8.2e11}$

$\approx 19ms$, which is too high. ~~But we can~~

So we have to increase our size to $4 \times 4 \times 4$. With this,

- ▷ our max $B \approx 290$. So we'll be compute bound & ~~memory~~ the
- ▷ linear ops. A lower bound on latency ~~is still~~ gives 14ms (by assuming
- ▷ ~~linear~~ there is bandwidth by training), so ~~the~~ 290 is too high.

- ▷ At $B=160$, we have a lower bound of 14ms, so

- ▷ we now need to check the FLOPs for linear.

- ▷ This is
$$\frac{2.405 \cdot 160 \cdot e9}{1.9724 \cdot 64} \approx 162ms$$
. This will not work
- ▷
$$\underbrace{1.9724 \cdot 64}_{\text{linear latency}}$$

- ▷ The latency for $W = \frac{160 \cdot 2.11}{5.2e11 \cdot 64} = 6ms$

- ▷ which gives total latency $> 162ms$, still too high.

- ▷ If we drop $B=140$, we get 9ms linear and are 5.6ms for kv
- ▷ loading, so we have 14.6ms total per iteration $< 15ms$.

- ▷ ~~This gives a rough idea of~~ Unfortunately, this means we

- ▷ we'd have to check 64 ways, but our 18kitt is $\frac{52248}{140 \cdot 8} = 4.7$.

- ▷ We could go to $4 \times 8 \times 4$ and do 2D model checks, $2.25 \times$ as fast

- ▷ which would give $t_{lin} = \frac{4BD}{2X} + \frac{4BD\alpha X}{N}$ where $\alpha = \frac{1}{2}$, so

- ▷ we have $\frac{4BD}{2X} + \frac{12.75BDX}{N}$. $X_{opt} = \sqrt{\frac{N}{2\alpha}} \approx 4$.

- ▷ so we have $\frac{1}{2}BD + \frac{50BD}{128} \leq BD$

$$\hookrightarrow t_{\text{fci}} \leq \frac{BD}{W_{\text{fci}}}, \text{ and } T_{\text{fci}} > T_{\text{fci}} + T_{\text{fci}}$$

$$\Leftrightarrow \frac{W_{\text{fci}}}{W_{\text{fci}}} > \frac{2F}{N \cdot B} \Rightarrow N > \frac{2F}{8 \cdot \beta} \cdot \text{form 5},$$

$\beta = 8$, so we have that for $\beta = 120$, $N > 128$.

Great! This means for $\beta = 120$, we can do 2D model parallelism and ~~not be~~ W_{fci} -band. This means latency $\approx \frac{(120 \cdot 2 \cdot 11 + 405) \cdot 6B}{128 \cdot 8 \cdot 2 \cdot 11} \approx 6 \text{ms}$, great!

$$\text{Throughput} = \left(\frac{1}{\text{latency}} \right) \cdot 120 = 19135 \text{ tokens/second.}$$

$$\text{Our profile latency is } \frac{8192 \cdot 2 \cdot 405 \cdot 9}{1 \cdot 97614 \cdot 128} = 263 \text{ms.}$$

so, Profile produces $\frac{P}{263 \text{ms}}$ seq per second, and generation

$$\text{consumes } \frac{BG}{6 \text{ms} \cdot 8192} \approx \frac{1}{4}, \text{ so } \frac{P}{263} = \frac{1206}{6 \cdot 8192},$$

(if we assume median depth = 8192)

$$\frac{P}{G} = 0.64, \text{ so } \frac{P}{G} \approx 1.5, \text{ so to be fully}$$

saturated, we'd like 3 generation servers and 2 profile servers

(assuming 8192 maxLen
prefill and
generation).

To conclude, we want to serve with

4x8x4 TRV slice with 2D-model shading
using 3 generation servers and 2 prefill servers.

This gives us a decode per frame latency = 6ms and
prefill seq latency of 263ms, and a decode throughput of
19135 frames/sec.